

HRSEP 2001/02 FINAL REPORT

Project Title: Chandindu River Salmon Enumeration Weir

Project Reference Number: #01 - YT - ST - 001

Habitat Restoration Contract: # F1528 - FO02 - 0040

PROPONENT INFORMATION

Organization Name: Yukon River Commercial Fishing Association (YRCFA)
Tr'ondek Hwech'in (THFN)

Contact Name, Title: Jake Duncan

Mailing Address: Box 1375 Dawson City, Yukon Y0B-1G0

Phone: (867) 993-6210 w

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Email: jduncan@yknet.yk.ca

Web Page: N/a

Please list the **partners** in this project and their roles. Include the organizations, governments, funding sources, or volunteer groups that helped get this project started and completed. Roles may include (but are not limited to): letters of support, financial support, volunteer labour, help with acquiring permits, administration, access to landowner property, etc.).

YRCFA – Financial and In-Kind support, volunteer labour and Administration

THFN – In-Kind support, volunteer labour, administration and financial assistance

Yukon Territorial Government (YTG) – Has provided Communications, Health and Safety equipment.

Yukon River Panel – Financial Assistance – Community Capacity Training

Alaska Department of Fish and Game (ADF&G) – Resistance-board weir design and construction – technical assistance.

DFO – Technical Assistance

HRSEP – Financial Support

Yukon Salmon Committee/H CSP – Habitat Steward, technical assistance, community capacity training, organization.

PROJECT INFORMATION

PROJECT PREPARATION

Project Rationale (Why was this project important?)

As the "Flagship" community-based project of the sub-basin restoration and enhancement plan, the Chandindu River Salmon Enumeration Weir has provided valuable and necessary information on salmon resources in one of the sub-basin's important watersheds. Previous to implementing an area-wide strategic stock restoration program based on incubation and outplanting, this important assessment work must be completed. This project has provided important information in support of the Sub-Basin's Restoration and Enhancement Plan.

Was a feasibility study or pre-assessment done for this project? Yes ☒ No ☐

If yes, please describe when and by whom.

This project was conducted in 1998, 1999, and 2000 by the YRCFA and the THFN, with financial assistance from DFO, the Yukon River Panel and YTG.

Did you receive DFO input on this project? Yes ☒ No ☐

Name, Title, and Location of DFO Contacts that helped you with this project:

Lana Miller, Resource Restoration Biologist, DFO - Prince Rupert
Pat Milligan, StAD, DFO - Whitehorse
Al von Finster, Resource Restoration Biologist, DFO - Whitehorse

PROJECT OBJECTIVES (FROM "APPENDIX A" OF YOUR CONTRACT)

Objective # 1 : Enumerate Chinook salmon on the Chandindu River to assess the availability of broodstock and provide baseline data for the future evaluation of success/failure from strategic stock restoration (will determine the feasibility of a local area strategic stock restoration program). Provide DFO with in-season mark recapture data weekly.

Was it achieved?: Yes ☒ No ☐

Details:

Chinook salmon were enumerated on the Chandindu River during weir operations from July 1st to Sept 8th. 89 Male and 40 Female Chinook salmon and a total of 29 chum salmon were counted. A minimum of 12 female Chinook salmon (according to DFO protocols, 30%) would have been available for broodstock collection, yielding an estimated 96,000 eggs for incubation. A high water event during the peak of the salmon migration unfortunately rendered the weir unable to accurately assess the maximum number of fish returning to the system, however, the system's escapement will be re-assessed with information from the weir's carcass recovery efforts (using marked:unmarked carcasses and run-timing). The recent high water events (2000, 2001) have illustrated the need for a resistance-board weir on this system. DFO was provided with weekly mark-recapture information for their in-season run abundance estimates, and, scale samples, DNA samples, and spaghetti tag information were collected.

Objective # 2 :

Sample Juvenile Chinook salmon to determine optimum target grow-out sizes based on natural sizes in this system (will contribute to determining the feasibility of a local area strategic stock restoration program). Enumerate/assessment of chum salmon utilizing the Chandindu River Watershed.

Was it achieved?:

Yes ☒ No ☐

Details:

Juvenile Chinook salmon were sampled weekly and information from previous years' operations were compiled. In order to effectively tag artificially raised salmon (coded wire tagging), the target grow-out size for fry is 2 grams. In nature, from the data we have collected to date, it seems that wild fry on average do not reach 2 grams until between the 28th and 30th Statistical week (Mid-point for period: between July 8th and the 22nd), or roughly, mid-July. This determination is very valuable in assessing the feasibility of a local area incubation facility and program. Chum salmon have been enumerated on the Chandindu River system. An early run of chum salmon has been identified on this river, which are possibly a small but important "summer chum" stock.

Objective # 3 :

Gain valuable community-based experience and build the community's capacity, further the goals of the Sub-basin Plan, foster conservation and stewardship ethics, and provide stewardship initiatives to local area people.

Was it achieved?:

Yes ☒ No ☐

Details:

Much experience has been gained during the course of this project, and as a result, the community's capacity to conduct this type of project is much greater. This experience is extremely valuable and crucial to future R&E Plans. In terms of furthering the Sub-basin R&E Plan, this project has greatly contributed by filling in some of the critical unknowns, such as: natural juvenile sizes vs. optimum grow-out sizes (i.e. release dates, growth regimes, thus: water temperature requirements during incubation...etc.), overall broodstock availability, and, the ability of local area people to conduct this type of project. This project, as mentioned earlier, is the "flagship" project of the proponents. As a result of this project, the Commercial Fishing Association and the First Nation are now working together in partnership on many conservation and stewardship initiatives in the area. By employing Commercial Fishers and First Nation people, stewardship initiatives have been provided.

Objective # 4 :

Gain watershed knowledge to support habitat protection efforts.

Was it achieved?:

Yes ☒ No ☐

Details:

Watershed knowledge has been gained during the course of this project. Valuable information regarding juvenile and adult salmon has been collected. Now that this knowledge has been gained, the Commercial Fisher and First Nation partnership, and other entities such as DFO, now has information to base future habitat protection efforts on.

Objective # 5:

N/A

Was it achieved?:

Yes ☐

No ☐

Details:

Objective # 6:

N/A

Was it achieved?:

Yes ☐

No ☐

Details:

→ If you have more than six objectives listed in Appendix A of your contract (Project Description), please use an additional page to finish listing your objectives and results.

PROJECT SUMMARY

Please provide an overview of your project below. If required, you may attach an additional sheet. Your summary should include:

- Methods used to accomplish your completed or partially completed objectives
- Whether the project proceed according to plan, or if you encountered unexpected difficulties
- Whether you ran over or under budget
- Whether the project took longer or less time than projected

(Project Summary)

Project planning, hiring and coordination commenced mid-June. The project coordinator and contractor, Mike Taylor (THFN), who was assisted by the Dawson Area Habitat Steward (YSC-HCSP) completed project hiring (via FN protocols) by month's end. The crew then set-up camp and assembling the weir at the beginning of July. The weir was operational 16hrs per day from July 1st to September 8th. During this time 129 Chinook salmon (89 M, 40 F) and 29 Chum salmon were enumerated. In-season mark re-capture data was forwarded to DFO weekly and many scale samples, DNA samples, and, tag recovery information was collected and recorded. A minimum of 12 female Chinook salmon (according to DFO protocols, 30%) would have been available for brood stock collection, yielding an estimated 96,000 eggs for incubation. Juvenile Chinook salmon were sampled weekly and information from previous years' operations were compiled. In order to effectively tag artificially raised salmon (coded wire tagging), the target grow-out size for fry is 2 grams. In nature, from the data we have collected to date, it seems that wild fry in this system on average do not reach 2 grams until between the 28th and 30th Statistical week (Mid-point for period: between July 8th and the 22nd), or roughly, mid-July. This determination is very valuable in assessing the feasibility of a local area incubation facility and program.

During the peak of the Chinook migration (July 31st-Aug 7th) a high water event damaged and breached the weir, negatively effected the data collection. However, post-season data analysis will include estimating escapements using marked to unmarked information collected in the carcass recovery. The weir was disassembled slightly earlier than scheduled in order to freight the damaged sections back to town in order to repair them. The project was completed on time with minor scheduling changes and within budget. A separate final report with photos, graphs and results will be available shortly.

PROJECT STATUS AND FOLLOW-UP

Please describe the current status of the project. Please comment on the following:

- Whether the project is complete
- Has the problem being addressed been solved? (see "Project Rationale" on page 3)
- Recommendations for future work

The Chandindu River Salmon Enumeration Weir project is completed for 2001. Proposals to the Yukon River Panel's Restoration and Enhancement Fund for 2002-03 have been developed and submitted. These proposals address some of the difficulties encountered, such as constructing a resistance-board weir to deal with the frequent high water events, and, they address future weir operations and plans (to be used for brood stock collection).

Recommendations for future work would include: the construction of a resistance-board weir for the 2002 season and "fish-tubes" for holding brood stock; hatchery training for local people; completion of the Dawson Area Incubation/Outplanting Facility Feasibility Study; and, weir operations in 2002-03.

PROJECT SUMMARY STATISTICS

The statistics you will provide below help us to determine the specific and overall achievements of the Habitat Restoration and Salmon Enhancement Program. A summary report will be completed in the winter of 2001/2002, and distributed to all previously funded proponents. This report will be available to all interested individuals and organizations.

PERSONNEL

Total # of persons trained (staff and volunteers):

8

STAFF

Number of persons employed: 4+1

Person-days of employment created: 232

VOLUNTEERS

Number of volunteers involved: 4

Total number of volunteer hours: 250

ACTIVITY TYPE

Species Addressed (check all that apply):

Coho ☐
Pink ☐
Chum ☒
Other ☐
(Specify):

Chinook ☒
Sockeye ☐
Steelhead ☐

Activity type (check all that apply):

Inventory & Mapping (See #2 below) ☐
Public Awareness (See #3 below) ☒
Stock Enhancement (See #1 below) ☐
Other (Specify): ☒

Stock Assessment (See #1 below) ☒
Habitat Restoration (See #4 below) ☐
Stewardship/Community Planning (See #3) ☒
Strategic Stock Restoration

QUANTIFIABLE RESULTS

1. For Stock Assessment and Enhancement Projects:

Number of Juvenile Salmon Enumerated: 958

Number of Adult Salmon Enumerated: 141

Number of Salmon marked/tagged or released: all

Other (specify):

2. For Mapping & Inventory Projects:

Was your data collected in accordance with the DFO-HEB Info. Management guidelines or Sensitive Habitat Inventory Mapping (SHIM) guidelines?

Yes ☐ No ☐

If yes, was it submitted in digital format?

Yes ☐ No ☐

Linear or square metres of area mapped or inventoried:

m ☐

m² ☐

3. For Stewardship/Community Planning and Public Awareness Projects:

Number of Public Presentations/Media Releases:

Number of Landowners Contacted:

Other (specify):

(First Draft and consultations on the Sub-basin Restoration and Enhancement Plan completed)

4. For Habitat Restoration Projects:

Habitat Addressed (Check as many as apply):

In-channel ☐

Off-channel ☐

Riparian ☐

Other (Specify): ☐

Lake ☐

Estuarine/Marine ☐

Fencing:

Stream length protected

metres

Stream area protected

(fence-to-bank width x length along stream):

Square metres

Riparian replanting:

Area replanted

Square metres

Number of trees/shrubs planted

In-channel habitat:

Stream area restored

Square metres

Off-channel habitat:

Stream area created/restored

Square metres

Estuarine habitat:

Area created/restored

Square metres

Lake habitat:

Area created/restored

Square metres

Fish Access:

Length of stream made available

metres

Area of habitat made available (square metres):

Square metres

Other (specify):

This information will help us to enter your project into the "Fisheries Project Registry", a joint Federal-Provincial database that summarizes fisheries projects in British Columbia and the Yukon Territory.

Chandindu River

FISS Watershed Code:
80070980000000000000000000000000

Chandindu River Watershed

☐ Fraser River
☐ Mackenzie River
☐ Columbia River
☐ Skeena River
☐ Nass River
☐ Stikine River
☐ Taku River
☒ Yukon River
☐ Other/Marine

Yukon River North Mainstem Sub-Basin

Dawson City, Yukon

North Western Yukon Territory

Longitude: 139°42'26.9"

Northing: 7127493.25

You may attach additional documentation to illustrate your project's results. (optional)

Brochure ☐
News clippings ☐
Other ☒

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July 22/02

FINANCIAL SUMMARY

HRSEP FUNDING SUMMARY

Since your HRSEP project began, you have provided interim financial statements and project updates with your invoices to receive installments of your funding. One final financial statement that summarizes all spending of HRSEP money is required for your final payment. Please enclose your financial statement and final invoice with this summary report.

Total received from HRSEP to date:	\$57,416.25 (final payment: \$6,362.57 not included)
Date:	November 15 th , 2001
Total HRSEP money spent and reported on financial statement form (please attach statement):	\$63,778.82 ✓
Total received from HRSEP after final invoice is paid:	\$63,778.82 ✓

HRSEP FUNDING DETAILS

Please enclose an HRSEP standard financial statement. These are available by mail or e-mail upon request from Elizabeth Leboe at (604) 666-8515.

It is not necessary to forward copies of individual receipts and invoices. As per the terms of your Habitat Restoration Contract, please retain these in your files for a minimum period of three years, as DFO reserves the right to audit all HRSEP projects.

OTHER CONTRIBUTORS

Contributions to the project's total budget may have been from other agencies or in-kind contributions from your own organization; please provide basic details below. Please estimate in-kind donations if not known exactly:

<u>Funding Source Details</u>	<u>Amount of funding (please note if funding is 'in-kind')</u>
YRCFA (in-kind, approx., w/equipment usage)	\$ 5,000.00
THFN (in-kind, plus interest on advanced funds)	\$ 1,500.00
YSC/HCSP (approx. in kind)	\$ 1,500.00
Yukon River Panel (funding Community capacity training course)	\$5,000.00
	\$

TOTAL \$13K

Yukon River Commercial Fishing Association
Box 1375
Dawson City, Yukon
Y0B-1G0

INVOICE # 0101

RECEIVED

JUL 04 2001

DATE: Wednesday, July 04, 2001

TO: Elizabeth Leboe, Community Liaison Biologist
Habitat Restoration and Salmon Enhancement Program
Fisheries & Oceans Canada
Habitat & Enhancement Branch
360 - 555 West Hastings Street
Vancouver, BC V6B 5G3

RE: Contract #: F1528 - FO02 - 0040
Cost Code #: 5B142 - 31 - - 167 - 4474 - 50818
Reference #: 01-YT-ST-001
Project Name: Chandindu River Salmon Enumeration Weir

DESCRIPTION	AMOUNT
July 2 nd , 2001 progress payment as per Habitat Restoration Contract	<u>\$20,000.00</u>
TOTAL:	\$20,000.00

PROJECT UPDATE:

The Chandindu River Salmon Enumeration weir crew has been mobilized and they have freighted all necessary gear to the weir site. The crew has also set up the remote camp and cleared trails. The crew has made and freighted 400 sandbags and has, to date, assembled 50% of the 140ft weir. The contractor has purchased all necessary safety gear and project supplies, and, the first weekly juvenile sampling has been conducted.

The weir is expected to be completely installed and "fish-tight" by July 5th. Migration-timing seems to be late this year and the first fish is expected to migrate through the weir around July 10th. All plans and target dates will be met and things are running smoothly.

FINANCIAL STATEMENT:

Please find enclosed (with email-attached) a financial statement current to invoicing date.

Yours truly,

2nd payment
Financials rec'd.

\$12K spent to July 4/01

Desc.	Sub Line	Ass't	Line #	Amount	Amount
5B142	30	167	4474	50818	20,000.XX
Contract # F1528 - FO02 - 0040				TOTAL	20,000.XX
Dated July 12/2001				July 12/2001	

Jake Duncan
(for the organization) YRCFA

Yukon River Commercial Fishing Association

Box 1375
Dawson City, Yukon
Y0B-1G0

INVOICE # 0102

DATE: Friday, July 20, 2001

TO: Elizabeth Leboe, Community Liaison Biologist
Habitat Restoration and Salmon Enhancement Program
Fisheries & Oceans Canada
Habitat & Enhancement Branch
360 - 555 West Hastings Street
Vancouver, BC V6B 5G3

RECEIVED
AUG 6 2001

RE: Contract #: F1528 - F002 - 0040
Cost Code #: 5B142 - 31 - - 167 - 4474 - 50818
Reference #: 01-YT-ST-001
Project Name: Chandindu River Salmon Enumeration Weir

DESCRIPTION	AMOUNT
July 20 th , 2001 progress payment as per Habitat Restoration Contract	<u>\$24,660.00</u>
TOTAL:	\$24,660.00

PROJECT UPDATE:

The Chandindu River Salmon Enumeration weir has been operating smoothly, enumerating to date 114 adult Chinook salmon, 34 of which were females. In terms of brood stock collection and DFO protocols, to date the crew could have taken roughly 88,000 eggs for restoration purposes. Juvenile sampling is also being conducted weekly to determine target grow-out sizes. The weir has experienced some difficulty due to unusually high water in the system, however, the environmental difficulties have only resulted in a few days' missed data.

FINANCIAL STATEMENT:

Please find enclosed (with email-attached) a financial statement current to invoicing date.

Yours truly,

Jake Duncan
(for the organization) YRCFA

3rd payment
\$57K spent to July 20/01

Resp.	Bus. Line	Allot.	Line Obj.	Project	Amount
5B142	310	167	4474	50818	24,660.XX
FO/Contract Number					OFF
F1528 - F002 - 0040					TOTAL 24,660.XX
Section 04 FAA					
Signature Elizabeth Leboe Date August 13/01					
for G. Loga					

